Goal: Informative prior

Binary data 1 = defective 0 = not defective

Historically $\mu = 0.05$ $\sigma^2 = 0.0025$

Prior: $\theta \sim \text{Beta}(a, b)$

$$\mu = \frac{a}{a+b}, \quad \sigma^2 = \frac{ab}{(a+b)^2(a+b+1)}$$

Solution:

$$\frac{a}{a+b} = 0.05 \qquad a = 0.05(a+b)$$

$$0.95a = 0.05b$$

$$\boxed{19a = b}$$

$$0.0025 = \frac{ab}{(a+b)^2(a+b+1)} = \frac{19a^2}{(20a)^2(20a+1)} = \frac{19a^2}{400a^2(20a+1)}$$

$$0.0025 = \frac{19a^2}{400a^2(20a+1)}$$

$$a^2(20a+1) = 19a^2 \qquad a \neq 0$$

$$20a+1 = 19$$

$$20a = 18$$

$$a = \frac{18}{20} = 0.9, \quad b = \frac{19 \cdot 18}{20} = 17.1$$

Prior: $\boxed{\theta \sim \text{Beta}(0.9, 17.1)}$

Binomial Distribution

$$p(y|\theta) = \binom{n}{y} \theta^{y} (1-\theta)^{n-y} \longrightarrow \text{Beta distribution}$$

<u>Lesson 1 of module 2</u>: prior $p(\theta) = 1$ for $\theta \in [0,1]$

<u>Lesson 2 of Module 2</u>: informative prior Beta (a,b)

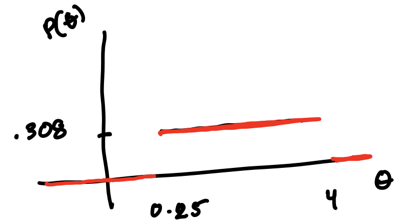
If our prior is a Beta distribution, what is the posterior?

$$p(\theta|y) \propto \overbrace{\theta^{y}(1-\theta)^{n-y}}^{\text{likelihood}} \; \overbrace{\theta^{a-1}(1-\theta)^{b-1}}^{\text{prior}}$$

$$= \theta^{y+a-1}(1-\theta)^{n-y+b-1}$$

$$\boxed{\theta|y \sim \text{Beta}(y+a,\; n-y+b)}$$

Prior :



$$P(\theta) = \begin{cases} .308 & \text{for } 0.25 \le \theta \le 4 \\ 0 & \text{o.w.} \end{cases}$$

posterior :

$$p(\theta|y) \propto p(\theta)\, p(y|\theta)$$

$$= \begin{cases} .308 \exp\left(-\frac{(y-\mu)^2}{2}\right) & 0.25 \le \theta \le 4 \\ 0 & \end{cases}$$

prior → uniform

posteriors → ??

likelihood → normal distribution

So this is not conjugate